

Assessing Credit Risk Using Logistic Regression

Domain: Retail Banking / Credit Risk | Tools: Binomial Logistic Regression (SPSS methodology) | Data: 500 Bank Loan Customers

Summary

Objective

Build a predictive model that identifies which customer characteristics are associated with loan default, so a bank can flag good versus bad credit risks before approving a loan.

Concepts / Methods Used

Binomial Logistic Regression, Wald Test (variable-level significance), Odds Ratio interpretation ($\text{Exp}(B)$), Cox & Snell / Nagelkerke Pseudo R^2 , Classification Table (predictive accuracy at 0.50 cutoff).

Key Output

- Sample: 500 bank loan customers (random sample drawn from 700 total cases per assignment spec); 26.2% historical default rate (131 of 500)
- Model fit: Nagelkerke $R^2 = 0.455$, Cox & Snell $R^2 = 0.311$ — the model explains a substantial share of the variation in default behavior
- Three variables are strong, significant predictors of default: Years with Employer ($p < .001$), Credit Card Debt ($p < .001$), and Years at Address ($p = .001$)
- Debt-to-Income Ratio is a weaker but still significant predictor ($p = .044$)
- Age, Education Level, Household Income, and Other Debt were NOT statistically significant predictors of default in this model
- Overall classification accuracy: 83.2% — but with an important asymmetry: 93.2% accurate at identifying non-defaulters, only 54.9% accurate at catching actual defaulters

Business Actionables

- Weight loan-approval scoring most heavily on employment tenure and existing credit card debt — these are the two strongest, most reliable red flags
- Treat short address tenure (frequent movers) as a secondary risk signal worth checking
- Do not rely on age, education, income, or 'other debt' as credit-risk filters — the data shows they don't meaningfully separate defaulters from non-defaulters here
- Because the model misses nearly half of actual defaulters (54.9% sensitivity), it should support human underwriting judgment, not replace it — treat it as a risk-flagging aid, not an automatic approval/rejection engine

Detailed Analysis — Model Output, Explained, With Actionables

Note: This assignment required performing a Binomial Logistic Regression in SPSS on a specific 500-case random sample of the bankloan dataset. The exact 500-case sample (already correctly filtered and matching the assignment's required sample size) was available in the submitted data file. The tables below reproduce that same statistical analysis — computed directly on this real, submitted dataset — to document the actual findings and provide portfolio-ready output; the same numbers would result from running this in SPSS on this sample.

Output 1 — Model Summary

-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
388.681	0.311	0.455

Model fit statistics: -2 Log Likelihood, Cox & Snell R², Nagelkerke R²

What this is: Standard logistic regression fit statistics that indicate how much of the variation in loan default the model explains, compared to a model with no predictors at all.

Layman meaning: Nagelkerke R² of 0.455 means the model captures a meaningful and fairly strong share of what separates defaulters from non-defaulters — for a real-world credit behavior model (which is inherently noisy), this is a solid result, not a weak one.

Actionable for the user: This model is strong enough to be used as a genuine decision-support tool for loan screening — not just a statistical exercise — though it should be paired with the Classification Table (Output 3) to understand exactly where it succeeds and fails.

Output 2 — Variables in the Equation

Variable	B	S.E.	Wald	df	Sig.	Exp(B)
Age	0.03	0.023	1.757	1	0.185	1.03
Education Level	0.066	0.153	0.185	1	0.667	1.068
Years with Employer	-0.263	0.039	44.496	1	0	0.769
Years at Address	-0.094	0.029	10.844	1	0.001	0.91
Household Income	-0.006	0.009	0.333	1	0.564	0.995
Debt-to-Income Ratio	0.069	0.034	4.06	1	0.044	1.072
Credit Card Debt	0.616	0.129	22.765	1	0	1.852
Other Debt	0.045	0.091	0.247	1	0.619	1.046
Constant	-1.453	0.762	3.636	1	0.057	0.234

Logistic regression coefficients, Wald statistics, significance, and odds ratios for every predictor

What this is: For each customer characteristic, this table shows its regression coefficient (B), the Wald test of whether that effect is statistically real (not chance), and the odds ratio (Exp(B)) — how much the odds of default multiply for each one-unit increase in that variable.

Layman meaning: Years with Employer has an odds ratio of 0.769 ($p < .001$) — each additional year with the same employer cuts default odds by about 23%, making job tenure a strong protective factor. Credit Card Debt has an odds ratio of 1.852 ($p < .001$) — every additional unit of credit card debt roughly makes default odds 85% higher, the single strongest risk driver in the model. Years at Address (odds ratio 0.910, $p = .001$) shows a similar, smaller protective effect — customers who've lived at the same address longer default less. Debt-to-Income Ratio matters too (odds ratio 1.072, $p = .044$) but more weakly. Age, Education, Income, and Other Debt all have p-values above 0.05 — none of them reliably predict default once the other factors are accounted for.

Actionable for the user: A loan officer building a quick risk checklist should ask: (1) How long has this applicant been with their current employer? (2) How much credit card debt do they already carry? (3) How long have they lived at their current address? These three questions carry almost all of the model's real predictive power — collecting age, education, or income data adds little extra value for this specific decision.

Output 3 — Classification Table

Classification Table (Cutoff = 0.50)

Observed	Predicted: No Default (0)	Predicted: Default (1)	Percentage Correct
No Default (0)	344	25	93.2%
Default (1)	59	72	54.9%
Overall Percentage			83.2%

Predicted vs actual default outcomes at the standard 0.50 probability cutoff

What this is: A comparison of what the model predicted (default or no default) against what actually happened historically, showing overall accuracy and the split between correctly and incorrectly classified customers.

Layman meaning: The model is very good at correctly identifying customers who will NOT default (93.2% accuracy on that group) but much weaker at catching customers who WILL default (only 54.9% accuracy) — it misses close to half of actual defaulters. The 83.2% overall accuracy figure looks strong but is misleading on its own, because it's inflated by the model's strength on the easier, majority group (non-defaulters make up nearly 74% of the sample).

Actionable for the user: A bank using this model as-is would still approve loans for a meaningful number of customers who go on to default. Before deploying this in production, the probability cutoff (currently 0.50) should be lowered to flag more borderline applicants as risky — trading some false alarms on safe customers for catching more real defaulters, since missing a defaulter is typically more costly to a bank than over-flagging a safe applicant.

Business Takeaway

Employment stability and existing credit card debt are the two dominant, reliable predictors of loan default — not age, education, or income, which is a genuinely useful (and slightly counter-intuitive) finding for credit policy design. The model is accurate overall but currently under-catches real defaulters, so it should be deployed as a risk-flagging aid alongside human underwriting judgment, with the decision cutoff tuned to catch more at-risk applicants before going live.